Closed-Orbit Tori and the Complete Original-Clock Spectrum of Compact Heisenberg Nilflows

HCS-C387 / HEN-O371

5 September 2026

Abstract

For the compact integer Heisenberg quotient, we study every member of the two-parameter family $W = X + \beta Y + \gamma Z$ without changing its physical clock. At irrational slope there are no closed orbits. At rational slope p/q, one invariant circle-valued phase gives every primitive period: a phase of reduced denominator d has least period dq, and time kq fixes exactly k connected two-tori. The return maps are clean and unipotent, so a finite isolated-orbit census cannot represent these continuous families. An explicit signed-mode Weil–Brezin transform gives the full Hilbert-space decomposition. Chirp conjugation identifies the exact self-adjoint domains, the toral pure-point component, and the countably infinite multiplicity Lebesgue component; no singular-continuous spectrum remains.

Keywords: Heisenberg nilflows; closed orbits; clean fixed tori; Koopman spectrum; Weil–Brezin transform; rational slope

中文摘要

本文在原物理时钟下研究紧海森堡商空间上的完整双参数流族。
无理斜率没有闭轨；有理斜率的一个不变相位给出全部本原周期，
固定集是连续二维环面而非孤立周期点，返回映射具有幂幺剪切。
显式正负中心模变换和啁啾共轭给出完整希尔伯特空间分解、
自伴定义域、纯点及勒贝格谱重数，并排除奇异连续谱。

关键词： 海森堡流；闭轨；干净固定环面；库普曼谱；
魏尔布雷津变换；有理斜率

1 The frozen source and the contribution

Let $H = \mathbf{R}^3$ with

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy'), \qquad M = \mathbf{Z}^3 \backslash H.$$

Haar measure has total mass one, represented by Lebesgue measure on the unit cube. The left-invariant fields are $X = \partial_x$, $Y = \partial_y + x\partial_z$, $Z = \partial_z$. We fix the coefficient of X to one and allow all $\beta, \gamma \in \mathbf{R}$ in $W = X + \beta Y + \gamma Z$. This is a declared family, not a normalization performed at a vanishing coefficient. Our conventions are $U_t f = f \circ \phi_t = e^{\mathrm{i}tA} f$ and $A = -\mathrm{i}W$.

Classical nilflow theory already distinguishes the toral factor from relative Lebesgue spectrum and mixing on its orthogonal complement; Avila–Forni–Ulcigrai explicitly recall these facts and the constant-roof skew-shift construction [1, Sections 2.1–2.2]. Flaminio–Forni place invariant distributions and cohomological equations in the broader nilflow setting [2]. We reconstruct the source results needed here directly, including rational slopes and exact operator domains. We do not claim novelty for the classical spectral or ergodic mechanisms, and do not transfer any theorem about nontrivial time changes to this unchanged flow. Within the present research sequence this is a continuous right-translation owner, not the discrete Heisenberg automorphism of C146/C151/C156, nor the noncompact sub-Riemannian geodesic of C270.

2 Every closed orbit and every fixed torus

Round 0 advance. A single source-invariant phase classifies all rational-slope primitive periods and exposes why an isolated-cycle determinant is not available.

The flow and the lattice identification are

$$\phi_t(x,y,z) = (x+t, y+\beta t, z+\gamma t+\beta xt+\beta t^2/2), \tag{1}$$
$$(x,y,z) \sim (x+r, y+s, z+k+ry), \qquad r,s,k \in \mathbf{Z}. \tag{2}$$

Equation (1) is right multiplication by $(t, \beta t, \gamma t + \beta t^2/2)$, so it descends to M, is complete and preserves Haar measure. The x velocity is one, hence no point is stationary. By (2), a nonzero return requires $t \in \mathbf{Z}$ and $\beta t \in \mathbf{Z}$. Thus irrational β has no closed orbit at any point or any nonzero time.

Theorem 1 (Complete rational-slope atlas)**.** *Let $\beta = p/q$ with $(p,q) = 1$ and $q > 0$, including $(p,q) = (0,1)$. The circle-valued function*

$$\theta(x,y) = \gamma q + px - qy + pq/2 \pmod 1$$

is well-defined and invariant on M. A point closes if and only if θ is rational. If its reduced denominator is d, its least positive period is dq. At time $kq > 0$ the fixed set is exactly k connected two-tori, and the set of points with least period dq is exactly $\varphi(d)$ connected two-tori, with $\varphi(1) = 1$.

Proof. Integer lattice changes add $pr - qs$ to θ; its derivative along W is $p - q\beta = 0$. Any horizontal return time is dq for an integer d. The central displacement after subtracting the required lattice action is

$$\gamma dq + \beta x dq + \beta(dq)^2/2 - dqy = d\theta + pq\, d(d-1)/2.$$

The last term is integral, so return is exactly $d\theta = 0$ in $\mathbf{R}/\mathbf{Z}$. This gives the least-period assertion, including $\theta = 0$. For time kq, the permissible phases are exactly j/k, $0 \le j < k$. The base map $px - qy : \mathbf{T}^2 \to \mathbf{T}$ has connected circle fibres because (p,q) is primitive. The central circle bundle restricts trivially to such a base circle, hence its preimage is a connected two-torus. More explicitly, lift the base circle on an interval; its endpoint central shift can be absorbed by a linear change of the central coordinate. Finally precisely $\varphi(d)$ phases have reduced denominator d. ☐

Each primitive torus contains continuum many distinct closed flow orbits, rather than a finite collection of primitive cycles. In local quotient coordinates the return derivative is

$$D\phi_{kq} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ pk & -qk & 1 \end{pmatrix}.$$

Indeed differentiating the lifted flow gives the entry βt in the x column, while undoing the lattice shift subtracts $t\,dy$ from the central coordinate. Its difference from the identity has rank one and kernel $\{p\,dx - q\,dy = 0\}$, exactly the tangent plane of the fixed torus. The fixed set is therefore clean. All multipliers equal one, and quotienting out the flow direction leaves a nontrivial unipotent two-dimensional Poincaré return. In particular $\det(I - D\phi_{kq}) = 0$; a sampled torus cannot supply a finite isolated-orbit denominator.

For a strobe $h \neq 0$ at rational slope, if h/q is irrational, no positive iterate returns anywhere. If h/q is rational, some positive iterate has an uncountable fixed set. Thus ordinary fixed-cardinality Artin–Mazur zeta functions are either one or undefined. Irrational slope always gives the former, for every nonzero strobe. This dichotomy does not compute a spectral determinant.

3 The entire Hilbert space, including negative central modes

Round 1 advance. The orbit atlas is supplemented by a complete unitary decomposition, exact domains and multiplicities; it is not a finite-mode approximation.

Expand centrally, $f(x,y,z) = \sum_{m\in\mathbf{Z}} e^{2\pi i mz} f_m(x,y)$. The quotient convention imposes

$$f_m(x+r, y+s) = e^{-2\pi i mry} f_m(x,y).$$

The $m = 0$ space is $L^2(\mathbf{T}^2)$. For $m \neq 0$, expand in y: $f_m(x,y) = \sum_k f_{m,k}(x) e^{2\pi i ky}$. The shift $x \mapsto x+1$ gives $f_{m,k}(x+1) = f_{m,k+m}(x)$.

Writing $k = j + m\ell$, where $0 \le j < |m|$, gives the unique coefficient representation $f_{m,j+m\ell}(x) = g_j(x+\ell)$. Consequently define

$$V_{m,j}g = e^{2\pi \mathrm{i} mz} \sum_{\ell \in \mathbf{Z}} g(x+\ell) e^{2\pi \mathrm{i}(j+m\ell)y}.$$

For Schwartz g this is a smooth lattice-invariant function; negative m changes the residue ordering but not the representation. Parseval in y and the tiling of $\mathbf{R}$ by $[0,1]+\ell$ give $\|V_{m,j}g\|_{L^2(M)} = \|g\|_{L^2(\mathbf{R})}$. Orthogonality of residues and the reverse coefficient construction prove surjectivity, hence

$$L^2(M) = L^2(\mathbf{T}^2) \oplus \bigoplus_{m \neq 0} \bigoplus_{j=0}^{|m|-1} V_{m,j} L^2(\mathbf{R}). \tag{3}$$

This proof identifies the entire space, rather than merely displaying some invariant subspaces.

Theorem 2 (Domains and complete spectrum)**.** *On the torus block, $e^{2\pi \mathrm{i}(kx+ly)}$ has eigenvalue $2\pi(k+\beta l)$. On each noncentral block,*

$$A_{m,j} = -\mathrm{i}\frac{d}{du} + 2\pi(\beta m u + \beta j + \gamma m).$$

Put

$$\psi_{m,j}(u) = \pi\beta m u^2 + 2\pi(\beta j + \gamma m)u, \qquad C_{m,j}g = e^{-\mathrm{i}\psi_{m,j}}g.$$

The exact self-adjoint domain is $C_{m,j}H^1(\mathbf{R})$. The toral spectral measure is pure point, each noncentral block has multiplicity-one Lebesgue spectrum, the total noncentral multiplicity is countably infinite, and there is no singular-continuous component. The spectrum of A is $\mathbf{R}$ for every β, γ.

Proof. Substitute the transform into $-\mathrm{i}(\partial_x + \beta\partial_y + (\gamma + \beta x)\partial_z)$. On the term indexed by ℓ, use $u = x + \ell$ to obtain the stated affine potential. Since $\psi'_{m,j} = 2\pi(\beta m u + \beta j + \gamma m)$, direct differentiation yields $A_{m,j}C_{m,j} = C_{m,j}(-\mathrm{i}\partial_u)$. Unitary conjugation of the momentum operator on $H^1(\mathbf{R})$ proves the domain assertion and self-adjointness. In particular the domain is not replaced by a separate requirement that both g' and ug lie in L^2; cancellation under the chirp is part of the actual domain.

The toral domain is the set of Fourier coefficients satisfying

$$\sum_{k,l} |2\pi(k+\beta l)|^2 |\hat{f}_{k,l}|^2 < \infty.$$

Combine this with the noncentral domains using square-summable graph norms. Schwartz functions after chirp multiplication form smooth block cores. On these dense cores the generated unitary group and the actual Koopman flow agree by integration of the first-order equation. Their extensions agree by unitarity, so the assembled operator is the original generator.

Fourier transformation of $-\mathrm{i}\partial_u$ gives multiplication by $2\pi\xi$ on $L^2(\mathbf{R})$, proving multiplicity-one Lebesgue spectrum. There are countably many blocks in (3). The torus contributes only its character eigenvectors, which are a complete orthonormal basis. No other spectral type can occur. ☐

For irrational β, the toral eigenvalues are distinct and dense in $\mathbf{R}$. For rational $\beta = p/q$, their set is $2\pi\mathbf{Z}/q$, and each has countably infinite multiplicity since $qk + pl$ takes every integer with infinitely many representations. These are assertions about the toral block; the full generator always also has its Lebesgue component.

A correlation in one noncentral block is the Fourier transform of an L^1 product of two L^2 Fourier transforms, so it tends to zero as $|t| \to \infty$ by the Riemann–Lebesgue lemma. Finite block approximation and Cauchy–Schwarz extend this conclusion to the whole noncentral subspace. The full flow is never weakly mixing, because the nonconstant toral eigenfunction $e^{2\pi \mathrm{i} x}$ survives for all parameters.

4 Scope and limitations

This is a theorem about a frozen continuous source, including zero and negative slopes and arbitrary central velocity, not a target spectral identification. Rational-slope periods use the same original clock as the irrational flow. The $X = 0$ family is outside scope. Classical prior art is acknowledged; no time-change mixing theorem or quantitative equidistribution rate is asserted. No target arithmetic local data, Euler factors, root number, automorphy, divisor, functional equation, zero match or Hilbert–Pólya operator is claimed. Route B remains disabled.

This work used AI-assisted proof exposition and exact-code development. Separate same-family agents read the proof and manuscript and tested additional hostile inputs; these are internal reviews, not external human review or formal verification. No unpublished material was uploaded to an external model.

References

[1] A. Avila, G. Forni and C. Ulcigrai, *Mixing for Time-Changes of Heisenberg Nilflows*, arXiv:1003.4636v3 (2010). `https://arxiv.org/abs/1003.4636`.

[2] L. Flaminio and G. Forni, *On the cohomological equation for nilflows*, Journal of Modern Dynamics **1**(1) (2007), 37–60. `https://doi.org/10.3934/jmd.2007.1.37`.